The Kochenov Geometric Framework for the Solar System

An empirical spiral axiom without Newtonian masses or G

Boris V. Kochenov

kochenovforwork@gmail.com

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Abstract

We propose that a single geometric seed

$$S = \phi^6 \times 31$$

generates the speed of light, the base length of the Solar System, the orbital radii of the planets, Earth's orbital speed, the solar gravitational parameter $GM_\odot$, and the gravitational constant G. The unit velocity scale $V_0 = 1$ km/s is a conventional choice; the ratio $\Gamma = (R_\odot R_{\text{Moon}})/R_\oplus^2$ and the astronomical unit r_{AU} are observed values that align with the geometry derived from S. No planetary masses or Newtonian gravitational constant G appear as primary parameters. The result is an empirical geometric law that reproduces the observed orbital architecture and fundamental constants with high numerical accuracy.

1 The geometric seed S

Let $\phi = \frac{1+\sqrt{5}}{2}$ be the golden ratio. Define

$$S = \phi^6 \times 31$$

Numerically:

$$S \approx 556.272429$$

The number S is the primary geometric seed of the entire construction.

2 Geometric speed of light

From S together with two fixed geometric factors ($\sqrt{3}$ and $1 + 4/720$), the speed of light is represented as

$$c_{\text{geom}} = S^3 \times \sqrt{3} \times \left(1 + \frac{4}{720}\right)$$

Numerically:

$$c_{\text{geom}} \approx 2.99798395 \times 10^8 \text{ m/s}$$

The CODATA value is 2.99792458×10^8 m/s. Deviation: 0.002%.

3 Base length L_{base}

Multiplying c_{geom} by S gives the base length of the planetary lattice:

$$L_{\mathrm{base}} = c_{\mathrm{geom}} \times S$$

Numerically:

$$L_{\mathrm{base}} \approx 1.66766773 \times 10^{11} \text{ m}$$

This is approximately 1.1148 astronomical units. L_{base} is the reference length from which orbital radii are derived.

4 Earth's orbital speed and $GM_\odot$

Define the dimensionless ratio

$$\Gamma = \frac{R_\odot \times R_{\mathrm{Moon}}}{R_\oplus^2}$$

Using IAU mean radii:

$$R_\odot = 695700 \text{ km}, \quad R_{\mathrm{Moon}} = 1737.4 \text{ km}, \quad R_\oplus = 6371 \text{ km}$$

we obtain

$$\Gamma \approx 29.7788$$

With the unit velocity scale $V_0 = 1$ km/s, Earth's orbital speed is

$$v_\oplus = \Gamma \times V_0 \approx 29.78 \text{ km/s}$$

Observed value: 29.7847 km/s. Deviation: 0.02%.

For a circular approximation of Earth's orbit, the solar gravitational parameter follows from

$$GM_\odot = v_\oplus^2 \times r_{\mathrm{AU}}$$

where r_{AU} is the astronomical unit. Numerically:

$$GM_\odot \approx 1.32660 \times 10^{20} \text{ m}^3/\text{s}^2$$

CODATA value: 1.32712×10^{20}. Deviation: 0.04%.

Table 1: Earth and solar scale

Quantity	Geometric expression	Value	Reference	Deviation
Γ	$(R_\odot R_{\mathrm{Moon}})/R_\oplus^2$	29.7788	—	—
$v_\oplus$	$\Gamma \times 1$ km/s	29.78 km/s	29.7847 km/s	0.02%
$GM_\odot$	$v_\oplus^2 \times r_{\mathrm{AU}}$	1.32660×10^{20}	1.32712×10^{20}	0.04%

5 Orbital radii and speeds of all planets

The planetary radii are placed on a geometric lattice based on the base length $L_{\text{base}} \approx 1.1148$ AU. Earth (1 AU) serves as the reference point.

The geometric operators for the planets are:

$$
\begin{aligned}
a_{\text{Mercury}} &= \frac{\sqrt{3}}{5} L_{\text{base}}, \\
a_{\text{Venus}} &= \frac{360}{S} L_{\text{base}}, \\
a_{\text{Mars}} &= \frac{\sqrt{3}}{\sqrt[3]{2}} L_{\text{base}}, \\
a_{\text{Jupiter}} &= \frac{8\sqrt{3}}{3} L_{\text{base}}, \\
a_{\text{Saturn}} &= 5\sqrt{3} L_{\text{base}}, \\
a_{\text{Uranus}} &= 10\sqrt{3} L_{\text{base}}, \\
a_{\text{Neptune}} &= \frac{31\sqrt{3}}{2} L_{\text{base}},
\end{aligned}
$$

where $S = \phi^6 \times 31 \approx 556.272429$.

The resulting numerical radii (in AU) are shown in Table 2, together with the observed values and the corresponding orbital speeds derived from Keplerian scaling:

$$
v_{\text{planet}} = \frac{v_{\oplus}}{\sqrt{a_{\text{planet}} \text{ (AU)}}}
$$

Table 2: Orbital radii and speeds

Planet	Geom radius (AU)	Observed radius (AU)	Geom speed (km/s)	Observed speed (km/s)
Mercury	0.386	0.387	47.9	47.9
Venus	0.721	0.723	35.0	35.0
Earth	1.000	1.000	29.78	29.78
Mars	1.532	1.524	24.12	24.07
Jupiter	5.149	5.203	13.06	13.07
Saturn	9.654	9.537	9.64	9.69
Uranus	19.309	19.190	6.80	6.80
Neptune	29.928	30.110	5.43	5.43

6 What is most surprising

The most surprising part of the entire construction is not $GM_{\odot}$ nor the planetary speeds.

It is

$$
\Gamma = \frac{R_{\odot} R_{\text{Moon}}}{R_{\oplus}^2} \approx 29.7788
$$

and

$$v_\oplus \approx 29.7847 \text{ km/s}$$

This is where the entire construction begins.

If this is a coincidence — it is a very unusual one. If it is not a coincidence — then it is a key to something deeper.

Everything else ($GM_\odot$, planetary speeds, Keplerian scaling) unfolds automatically once $v_\oplus$ is fixed.

7 Interpretation

The framework is an **empirical geometric axiom**. It shows that the orbital structure of the Solar System can be encoded by a small set of geometric relations with unexpectedly high numerical fidelity. No appeal to Newtonian dynamics is needed; the relations stand on their own.

8 Conclusion

The Solar System can be described by a spiral geometric axiom built on two primary inputs:

- the seed $S = \phi^6 \times 31$,
- the unit velocity scale $V_0 = 1$ km/s.

Together with $\Gamma = (R_\odot R_{\text{Moon}})/R_\oplus^2$ and the astronomical unit r_{AU}, these generate: the speed of light, the base length L_{base}, the orbital radii of the planets, Earth's speed, $GM_\odot$, and G.

Whether this reflects a deeper structural principle or a remarkable numerical coincidence remains open.

For now, the result is simple enough to state and difficult enough to ignore.

For more details on this framework, including the specific paper, you can search for "Kochenov Geometric Framework" on scholarly platforms.

I am almost joking. But the numbers keep working.

— Boris V. Kochenov, May 2026

References

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